

SahakarSetu (सहकार सेतु) | Technical Approach & System Architecture

Cooperative-Owned Digital Service Marketplace for Verified Skilled Trades (Electricians, Plumbers, Caregivers, Carpenters)

1. Full-Stack & Spatial Geo-Dispatch

PWA + PostGIS

- **Client Tier:** Offline-first Progressive Web App (PWA) built with React 18, Vite & Tailwind CSS. Low-bandwidth optimized for field craftsmen.
- **API Tier:** High-throughput Node.js/Express REST micro-router architecture with JWT session authentication.
- **Spatial Geo-Matching:** PostGIS spatial indexing (`ST_DWithin` / Haversine) for **<30 min emergency SOS dispatch** within a 3 km hyperlocal radius.

2. Fair-Wage Escrow Split (88 / 6 / 3 / 3)

0% Aggregator Cut

- **88% Direct Worker Payout:** Instant BharatQR / UPI settlement directly to worker's bank account upon completion (vs. 25-35% cuts on private apps).
- **6% Social Security & Welfare:** Automatically funds **PMSBY (₹2 Lakh accidental policy)** & continuous micro-pension ledger (+ ₹25/job).
- **3% Primary Co-op Society:** Strengthens local society working capital & tool loans.
- **3% Platform Infra:** Covers server hosting, SMS, maps & voice APIs.

3. Two-Tier KYC & Dual-OTP Protocol

DigiLocker + NSDC

- **Tier-1 Identity Check:** Automated DigiLocker Aadhaar/PAN validation with masked IDs (`XXXX-XXXX-1234`) compliant with the DPDP Act.
- **Tier-2 Skill Validation:** Skill India / NSDC qualification certification validated by Primary Cooperative Federation before dispatch activation.
- **Cryptographic Dual-OTP:** 4-digit **Start OTP** on worker arrival + completion verification eliminates ghost orders and dispute friction.

4. AI Surge Forecasting & Voice AI

LightGBM + Web Speech

- **Predictive Demand Engine:** LightGBM time-series model correlates meteorological telemetry (heatwaves >42°C, monsoon storms) with repair demand.
- **Workforce Mobilization:** Proactively reallocates cooperative technicians and issues transit vouchers to high-demand clusters.
- **Multilingual Speech AI:** Web Speech API provides hands-free audio job alerts in **Hindi & English** with one-tap voice acceptances (`"स्वीकार करें"`).

Key Impact: 88% Direct Worker Wage Floor Guaranteed

✓ ₹2,00,000 PMSBY Accidental Insurance

✓ <30 Min SOS Rapid Dispatch

✓ Zero Commercial Aggregator Price Gouging

100% Co-op Owned

System Architecture & Data Engineering Pipeline

End-to-End Flow from Citizen Booking to Worker Dispatch, OTP Verification, and Automated Escrow Settlement

01

Presentation Layer

Multi-Persona Progressive Web App

- **Citizen Portal:** Transparent itemized rate card catalog, 30-min SOS toggle, live worker radar & BharatQR checkout.
- **Shramik Worker PWA:** Web Speech AI voice assistant, live dispatch radar, Start OTP entry & verifiable Digital ID QR pass.
- **Federation Admin:** LightGBM demand heatmap, DigiLocker KYC queue & minimum wage rate governance.

02

API & Business Engine

Node.js / Express REST Services

- **Fair-Wage Calculation Engine:** Locks immutable 88/6/3/3 financial split server-side to prevent client tampering.
- **Dual-OTP Security Gate:** Enforces start & finish handshakes, updating status to `IN_PROGRESS` & `COMPLETED`.
- **Welfare Ledger Engine:** Auto-deposits 6% into PMSBY insurance policy and micro-pension fund per job.

03

Persistence & National Stack

Prisma ORM & PostGIS

- **Relational Relational Schema:** ACID-compliant models for Users, Workers, Bookings, Welfare Ledgers & Disputes.
- **PostGIS Spatial Querying:** Sub-second geospatial range queries (`ST_Distance_Sphere`) for proximity dispatch.
- **National Digital Stack Hooks:** Direct integration readiness for DigiLocker, Skill India (NSDC), and UPI 2.0 AutoSplit.